eLife

RESEARCH ARTICLE

How accurately can one predict drug binding modes using AlphaFold models?

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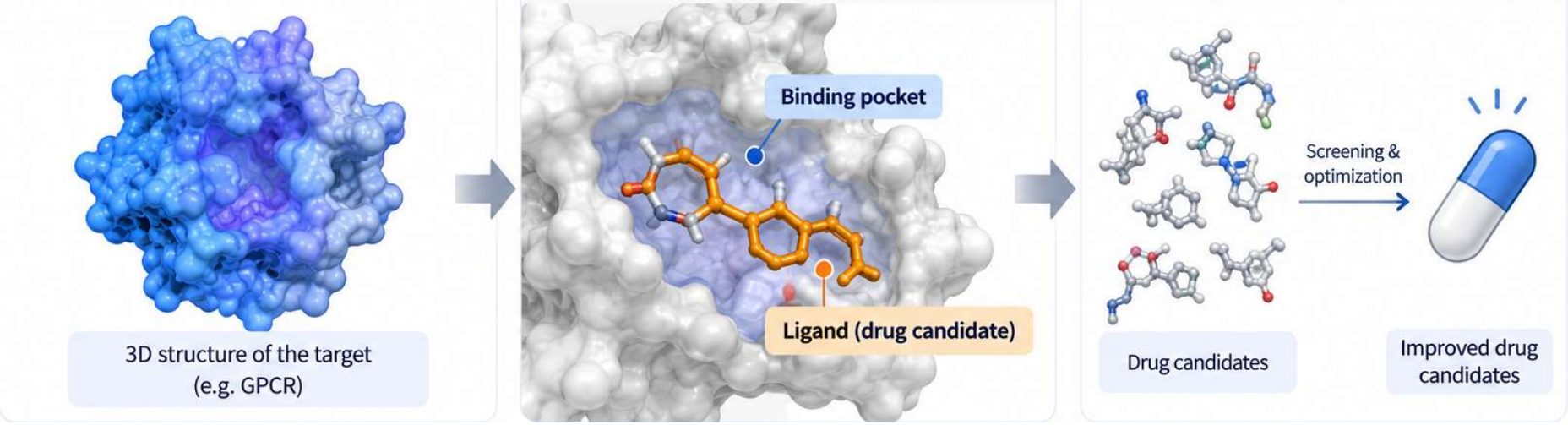
Structure → Drug Discovery

- Structures of the protein bound to many ligands

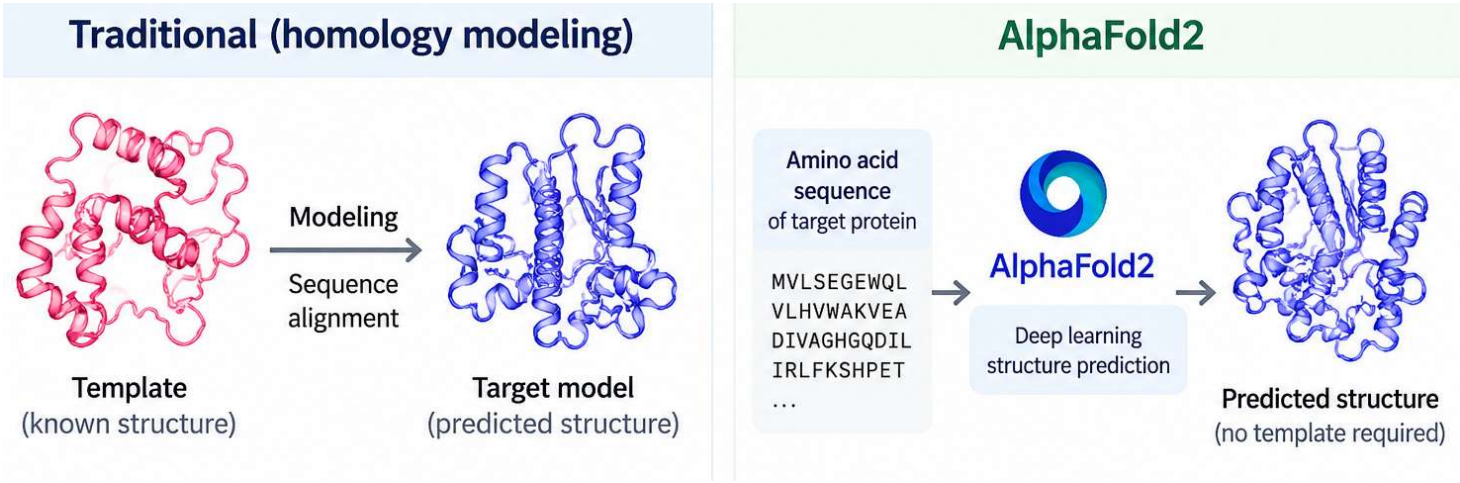
→drug candidates, drug optimization

- Structure – experimental methods (slow, expensive, not all structures available)

→ Computational methods



Traditional Modeling vs. AlphaFold2 – Protein Structure Prediction

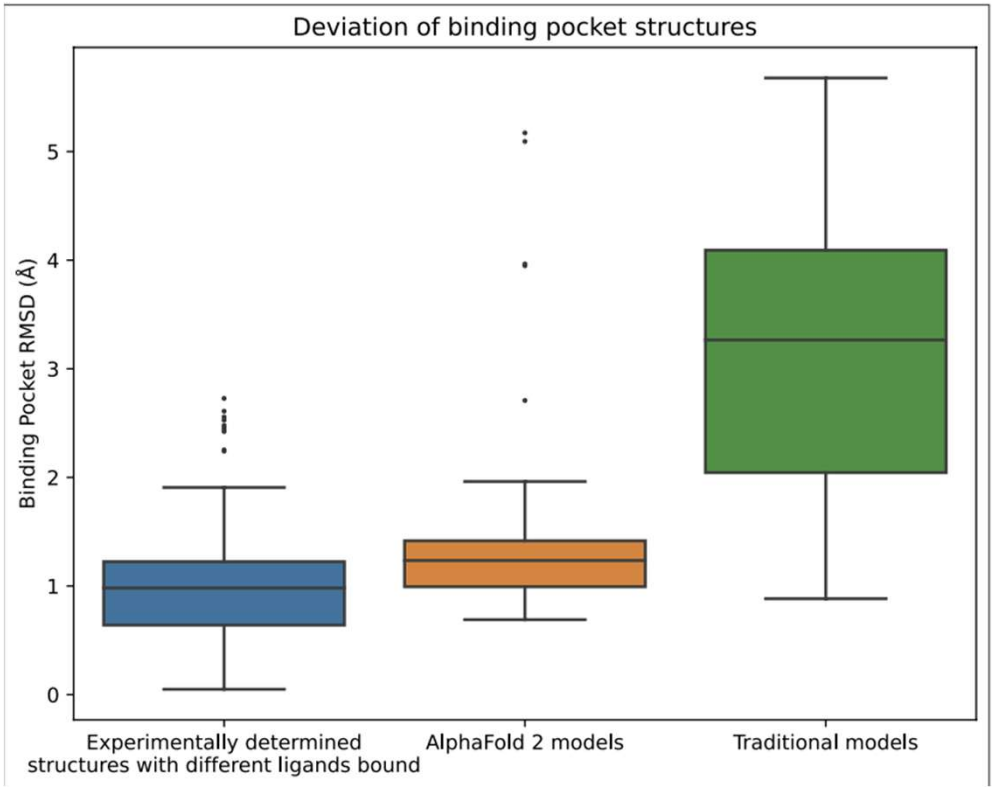


Q: To what extend can we utilize AF2 in predicting protein-drug interaction?

1) Binding pocket accuracy
2) Ligand binding pose accuracy

For a fair comparison, GPCRs which were experimentally determined after April 30th 2018 and traditional models that were published before April 30th 2018 were used.

Result – AF2 predicts binding pocket structure better than traditional models



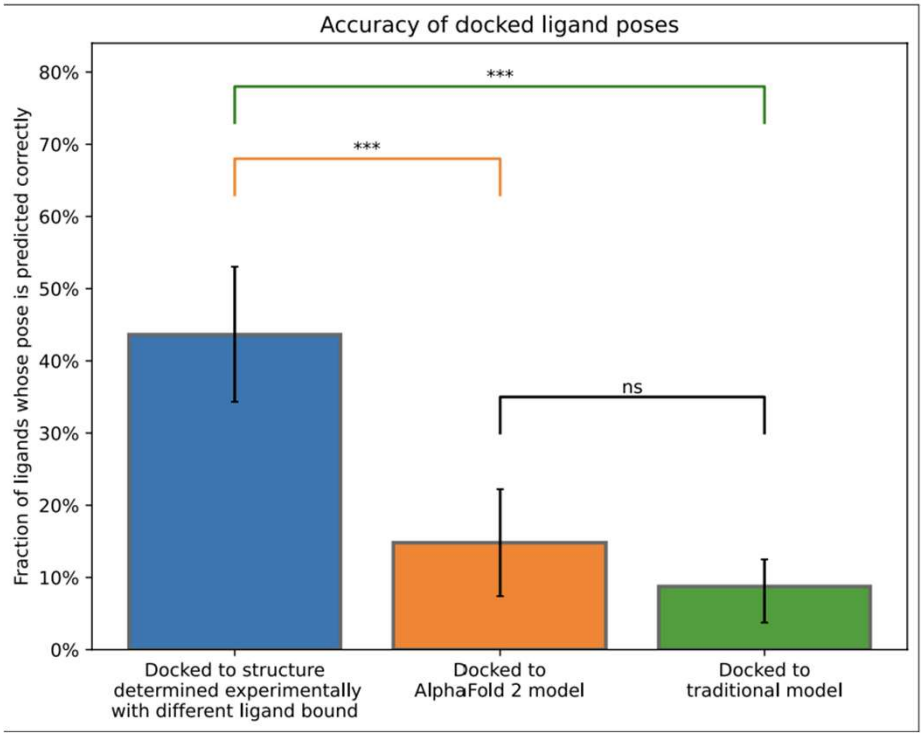
RMSD: root mean squared deviation

Measure how different two 3D structures are based on the positions of corresponding atom

Low – high accuracy

High – low accuracy

Result – Better structure accuracy didn't lead to better ligand binding pose



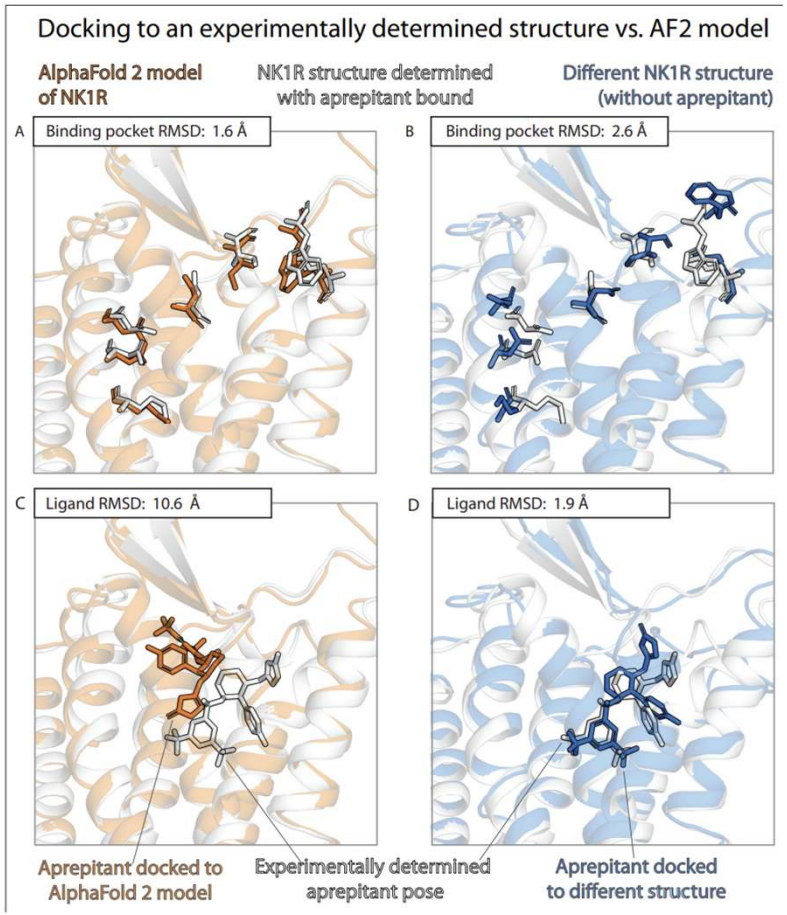
Computational docking

- Glide run in SP mode
- Glide run in XP mode
- Rosetta docking

AF2 & Traditional model

- Not statistically significant

Result – Better structure accuracy didn't lead to better ligand binding pose



Connection to our project

AF2 high structure accuracy doesn’t guarantee accurate DTI

Research Question 3: AlphaFold 정보는 어떤 난이도 구간에서만 기여하는가?
- AF2 high confidence가 prediction 과신을 유발할 수 있음

DTI 문제 유형	AlphaFold 구조를 추가했을 때
익숙한 target + 높은 구조 confidence	정확도 변화가 거의 없음
새로운 target + 높은 구조 confidence	정확도가 향상됨
새로운 target + 낮은 구조 confidence	정확도가 향상되지 않거나 악화됨